



Hand Dynamometry: The Essential Guide

Grip strength is rapidly emerging as an indispensable biomarker of health, with thousands of studies highlighting its relevance across various clinical specialties. It holds tremendous value for healthcare providers and is being integrated into leading clinical practices internationally.

However, there is a barrier to easy adoption: a gap between scientific knowledge and practical application. Literature supports the unquestionable importance of grip strength, but there is a lack of clarity on how to effectively incorporate it into clinical workflows.

This document provides guidance for unlocking the power of grip strength assessments as part of standard care.

We've distilled the most respected research on hand dynamometry to provide clear, actionable steps to choose the correct hand dynamometer and integrate grip strength measurements into everyday practice.

Step 1: Understand the Basics

Hand dynamometry is the measurement of grip strength. A clinician will use a hand dynamometer to measure how hard a person can squeeze to evaluate muscle function, monitor changes over time, and assess the risk of conditions like frailty, sarcopenia (muscle loss associated with age), or other health concerns. [1]

There are a few key concepts in hand dynamometry that are important to understand. These concepts will help you choose an appropriate hand dynamometer for clinical use, as well as apply research findings to your patient population.

The below concepts are aligned with research by Elaine Ewing Fess, a leading voice in defining the components of a clinically appropriate hand dynamometer and published in the Journal of Hand Therapy. [2]

a) Measurement Reliability

Measurement reliability refers to the consistency or stability of hand dynamometer readings over time without random or systematic errors. If a measurement is reliable, it will produce the same results under consistent conditions. Reliability and validity are quantified with an Intraclass Correlation Coefficient (ICC) score between 0 and 1.

There are a few requirements that must be met to demonstrate device reliability:

- **Accuracy:** The accuracy of a device can be impacted by both the sensor itself, as well as administration protocol, correct calibration, and the clarity of the readout.

Best Practice: Lower error is better.

- **Interrater Reliability:** The consistency of grip strength measurements when administered by different individuals (or raters). The interrater reliability of the product is impacted not only by the rater but also by the sensor and administration protocol.

Best Practice: Any score greater than 0.75 is considered good agreement and above 0.9 is considered excellent agreement

- **Test-Retest Reliability:** The consistency of grip strength measurements when taken at different time points using devices of the same brand.

Best Practice: Any score greater than 0.75 is considered good agreement and above 0.9 is considered excellent agreement.

b) Inter-Instrument Reliability

The consistency of measurements when compared to gold standard instruments with established reliability. It can be affected by differences in shape, sensor type, weight, and materials.

Best Practice: Any score greater than 0.75 is considered good agreement and above 0.9 is considered excellent agreement. An ICC score of 1 would never be expected between different products.

c) Sensitivity

The dynamometer's ability to detect small changes or differences in grip strength, and the clarity of the sensor's display in smaller increments.

Highly sensitive dynamometers can be used with a wider range of patients, especially those with extremely weak grip strength, such as older patients or individuals with neurological impairments.

Best Practice: The smaller the measurement increments, the better, reducing the risk of rounding errors that could obscure subtle changes in grip strength.

d) Administration Protocol

Standardized procedure for utilizing the hand dynamometer to take a Single Maximum Grip Test (SMGT) assessment, including position, verbal instructions, and interpretation of the results. Administration protocols are necessary to maintain the validity and reliability of the hand dynamometer, as they enable consistent use.

Best Practice: Protocols for administration must be developed in a way that allows for easy adherence and made available to all users.

e) Normative Data Set

A collection of standardized grip strength measurements from a large, diverse population. This data serves as a reference to compare an individual's grip strength against what's typical or expected for people of the same age and biologically assigned sex.

Best Practice: A published normative data set that reflects the current population.

f) Bibliography

A published set of research demonstrating all of the above requirements outlined. The purpose of the bibliography is to publish and share information on all the above criteria so that clinicians can make their own decisions and comparisons.

Best Practice: Research completed and made available to users illustrating the development and testing of the hand dynamometer.

Step 2: Choose the Right Product

The clinical hand dynamometer market features a wide variety of devices, and most do not provide measurement reliability.

The three primary categories of hand dynamometers are hydraulic, pneumatic, and mechanical devices, with digital and Bluetooth versions also gaining popularity.

Jamar hand dynamometers (Jamar and Jamar PLUS+) are the most widely used hand dynamometers on the market, up until this point considered the gold standard due to reliability and familiarity.

The Jamar / Jamar PLUS+ and GripAble hand dynamometers stand out for their rigorous scientific validation, thorough testing and extensive peer-reviewed research. As a result, we believe these are the only devices with sufficient evidence to be consistently recommended for clinical best practice, setting them apart in terms of accuracy, reliability, and clinical relevance.

GripAble vs Jamar

In the following section, we explore the similarities and differences between GripAble and Jamar, and lay out how they compare against each other in the clinical and operational characteristics previously described.

a) Included Components:

- **GripAble Device:** The electronic GripAble sensor is accompanied by the Able Assess platform which consists of a mobile app and an online portal. The original Jamar hand dynamometer is a hydraulic (non-digital) product with a visual gauge. The company has also released the Jamar PLUS+ that has a digital sensor and LCD screen to display readings.

Implication: The Able Assess platform features both an app and a portal, offering a comprehensive administration protocol that guides users—clinicians and patients alike—through a step-by-step process for completing grip tests. It also excels in data visualization, allowing users to compare their grip measurements to live population norms and track changes in grip strength over time. In contrast, Jamar products only display readings on the screen, which limits their data visualization capabilities.

b) Measurement Output:

- **GripAble Device:** The force measured by GripAble is consistent, providing a highly accurate and reliable output. The force output is different from Jamar, but this is normal (all dynamometers differ in output due to difference in size/shape/weight). On average, the force measured by the GripAble hand grip amounts to 69% of that measured by the Jamar PLUS+ for the same person. For instance, if the Jamar PLUS+ reads 10 kg, the GripAble will read approximately 6.9mkg. [3]

Implication: This difference is consistent for every reading, thus GripAble is a valid and reliable tool for grip strength measurement, meeting Fess's criteria for evaluating measurement instruments. The factors below contribute to the measurement differences between the GripAble and Jamar devices.

Characteristic	Best Practice	Jamar (published by Jamar)*	GripAble (including the Able Assess app)
Measurement Reliability: Accuracy	Lower variance is better	+/- 2 kg (+/- 4.4 lbs)	+/- 1.67 kg (+/- 3.7 lbs) (95th percentile) [3]
Measurement Reliability: Interrater Reliability	Pearson Correlation $\rho > 0.8$ is very strong*** ICC > 0.9 or excellent	Pearson Correlation $\rho = 0.996$ [4]	ICC = 0.991 (95% CI 0.98-1.0) [5]
Measurement Reliability: Test-Retest Reliability	Pearson Correlation $\rho > 0.8$ is very strong*** ICC > 0.9 or excellent	Pearson Correlation $\rho = 0.882$ [4]	ICC = 0.97 (95% CI 0.94 - 0.98) [6]
Inter-Instrument Reliability	ICC > 0.9 or excellent	Gold standard	ICC = 0.906 in comparison to Jamar PLUS+ (95% CI 0.87 - 0.94) [7]
Sensitivity	Smaller intervals are better	0.5 kg (1.1 lbs)	<100 g (.22 lbs) [3]
Equipment, administration, scoring	Protocols easy to use and available	Protocols published in the paper User Guide	Included within app
Normative Data Set	Published and comparable to modern populations	Originally published in 1985 and published in Jamar user guide.** [8]	UK: published in 2024 [5] US: to be published shortly
Bibliography	Published and readily available	Available, but not compiled in a list by Jamar	Available here [link]

*For this comparison, we assume that the Jamar and Jamar PLUS+ devices are equivalent. Research papers suggest that there is some variability between the Jamar and Jamar PLUS+ [9-13] however, Jamar itself has not published its own comparison study. Despite using different sensing mechanisms, the Jamar and Jamar PLUS+ have the same shape and size. Furthermore, Jamar continues to support the use of their normative data with the Jamar PLUS+.

**There have been revised norms published since then, but they have not been widely adopted and are not distributed in the Jamar User Guide. [8]

***Some older studies use the Pearson Coefficient Measure. A strong correlation of $r > 0.99$ demonstrates excellent agreement across measurements, to simplify reporting without separate values for left and right sides. The Pearson correlation coefficient measures the strength of a linear relationship between two variables (quantified as a score between -1 and 1), while the ICC is a measure of agreement for continuous outcomes.

c) Size:

- **GripAble Device:** Bigger circumference (141 mm) compared to Jamar (128 mm) in standard position 2, in an isometric mode.

Implication: A higher grip force contribution generated by the distal interphalangeal joints relative to metacarpal joints, a better reading for functional output, but leading to a lower overall force measured.

d) Shape:

- **GripAble Device:** Designed with a wider and rounder shape for consistent and comfortable placement during measurement, allowing users to exert their maximal effort.

Implication: The ergonomic “C” shape provides enhanced support for the webspace and arches of the hand, promoting accuracy by encouraging a consistent grip position.

e) Material and Weight:

- **GripAble Device:** Crafted from light-weight plastic (weighs 240 g) compared to the heavy metal-based Jamar (weighs 490 g).

Implication: Despite the difference in material, the lighter GripAble device is engineered and tested to endure the same force and a higher level of durability (repeated grip forces up to 90 kg) and withstand substantial drop impacts, ensuring reliability [4].

The GripAble Standard



FDA Registered as a Class II 510 k-exempt medical device



HIPAA compliant for data protection



Rigorously tested, with data published in multiple peer-reviewed scientific journals [\[see here\]](#).

Step 3: Implement!

Even the most clinically appropriate dynamometer that meets Fess's criteria may not align with the fast-paced, data-driven demands of modern healthcare systems. Research with clinicians across various specialties has identified several key challenges in operationalizing grip strength as a benchmark in clinical settings.

The following outlines the main challenges clinicians face when using grip strength as a biomarker in practice, along with solutions provided by GripAble and the Able Assess Platform.

a) Calibration

Leading hand dynamometers on the market require periodic returns to the manufacturer for recalibration. For example, Jamar recommends recalibration every 12 months, which incurs significant costs and necessitates the device's absence from the clinic for weeks.

The GripAble and Able Assess Solution: The GripAble sensor design ensures reliable long-term performance while minimizing the need for frequent recalibration [4]. Rigorously bench-tested, the GripAble sensor has demonstrated its ability to maintain accuracy and calibration even after multiple drops. Additionally, it includes an in-app function that allows clinicians to quickly reset (tare) the sensor to zero for immediate adjustments when needed.

b) Recording and Reporting

Standard protocol for measuring grip strength requires clinicians to take three measurements on each hand and calculate the average. Manually recording these measurements and comparing them to normative tables or previous results can be time-consuming.

The GripAble and Able Assess Solution: The Able Assess platform streamlines the assessment process by automatically performing calculations. It provides percentile ranks stratified by age and biological sex, along with mid-range averages for similar individuals. The app allows for a single grip strength test, saving time while maintaining good to excellent test-retest reliability (ICC = 0.91, 95% CI 0.83-0.95 for one trial; ICC = 0.97, 95% CI 0.94-0.98 for three trials) [7]. Additionally, the platform integrates with health records for immediate documentation and efficient outcome comparisons over time.

Single Maximum Grip Test Explained

Measuring grip strength involves administering a Single Maximum Grip Test (SMGT), which assesses peak grip strength by recording the maximum force exerted in a single grip. The Able Assess App includes built-in testing guidelines recommended by the American Society of Hand Therapists (ASHT), detailing testing positions and instructions [1].

Able Assess allows for either one or three repetitions, the option for a single measurement acknowledges time constraints in clinical practice. A test-retest reliability study by McGee [7] demonstrated good to excellent reliability for a single trial (ICC of 0.91, 95% CI 0.83 - 0.95), comparable to three trials (ICC = 0.97, 95% CI 0.94-0.98).

For three measurements, the highest value will be used; for a single repetition, the result will determine the patient's percentile score based on normative reference values stratified by age and biological sex. The app also provides the average grip strength range for patients of the same age and biological sex.

c) Hand Dynamometer Differences

Published research has utilized various hand dynamometers. Findings from studies using one dynamometer may not be applicable to clinical practice with another, complicating the understanding of relevant results.

The GripAble and Able Assess Solution: A comparative study between GripAble and the Jamar PLUS+ (the most widely used hand dynamometer [3]) now enables findings from studies conducted with Jamar or Jamar PLUS+ to be relevant for populations using the GripAble sensor.

Additionally, a normative data set has been published, enabling the application of percentiles from other publications directly to this data set.

Quick reference sheets for various populations and conditions where grip strength is significant have also been published [link](#), converting findings into equivalent GripAble measurements.

	Jamar	Jamar PLUS+	GripAble (including the Able Assess app)
Sensor	Hydraulic	Digital	Digital
Measurement Display	Visual Gauge	LCD Screen	In-App or on Portal
Calculating Averages	Manual	Automatic	Automatic
Record Keeping	Manual	Manual	Can be transferred by API to EHR
Tracking Measurements Over Time	Manual	Manual	Can be transferred by API to EHR
Calibration	Must be mailed in every 12 months or whenever dropped	Must be mailed in every 12 months or whenever dropped	Measurement accuracy robust to long-term/multi-year use and dropping.

Note: Conversion of study findings from Jamar to equivalencies with the GripAble sensor are made possible through a comparison study performed with many subjects [3]. While it is appropriate to convert grip strength values that have been calculated at the population-level e.g. risk-based cut-offs, these equivalencies should not be used on an individual basis e.g. interchangeable device use for a single individual.

Cut-Off Points and Quantiles Explained

In order to understand the best ways to apply grip strength research to a patient population, it is helpful to understand cut-off points and quantiles.

Cut-Off Points

Cut-off points are single measurements that help determine if a patient is in the “red zone,” indicating high risk. These are especially useful for quick assessments during routine visits or when the patient presents with other conditions. See various resources as grip strength as a biomarker.

Note: Conversion of study findings from Jamar to equivalencies with the GripAble sensor is made possible through a comparison study done with many subjects [3]. While this is appropriate to do at a population level, these equivalencies should not be used on an individual-to-individual basis.

Quantiles

Quantiles show an individual’s grip strength relative to peers, using age and sex-specific benchmarks from normative data to provide a percentile ranking. For instance, being in the 1st quantile (0-25th percentile) indicates lower grip strength than 75% of similar peers.

Clinicians should monitor quantiles to assess patient health over time, particularly noting significant drops into lower quantiles that persist for days or weeks. Such declines warrant further investigation, considering the patient’s comorbidities and risk profile.

References

- [1] Shechtman, O. & Sindhu, B. S. **American Society of Hand Therapists TM Grip Strength Key Recommendations for Outcome Evaluation of Grip Strength.** [\[Link\]](#)
- [2] Fess, E. E. . **Guidelines for evaluating assessment instruments.** Journal of Hand Therapy : Official Journal of the American Society of Hand Therapists, 8(2), 144–148.(1995). [\[Link\]](#)
- [3] Mutalib, S. A., Mace, M., Seager, C., Burdet, E., Mathiowetz, V., & Goldsmith, N. **Modernising grip dynamometry: Inter-Instrument reliability between GripAble and Jamar.** BMC Musculoskeletal Disorders, 23(1), 80. (2022). [\[Link\]](#)
- [4] Mace, M., Mutalib, S. A., Ogrinc, M., Goldsmith, N., & Burdet, E. . **GripAble: An accurate, sensitive and robust digital device for measuring grip strength.** Journal of Rehabilitation and Assistive Technologies Engineering, 9(1), 20556683221078456.(2022). [\[Link\]](#)
- [5] Mathiowetz, V., Weber, K., Volland, G., & Kashman, N. . **Reliability and validity of grip and pinch strength evaluations.** The Journal of Hand Surgery, 9(2), 222–226. (1984). [\[Link\]](#)
- [6] Mutalib, S. A., Sharma, D., Pike, S., Gwynne, L., Hyde, S., Morehouse, J., Davey, H., Edwards, L., Douglass-Kirk, P., Burdet, E., Goldsmith, N., & Mace, M.. **GripAble: Interrater reliability and normative grip strength of UK population.** Journal of Hand Therapy. (2024). [\[Link\]](#)
- [7] McGee, C.W., **Test-Retest Reliability and Precision of GripAble**, A Multi-Purpose Exergaming (under review)
- [8] **Jamar_Plus_Hand_Dynamometer_Instruction_Manual_012221.** (n.d.). [\[Link\]](#)
- [9] Cooper, R., Lessof, C., Wong, A., & Hardy, R. . **The impact of variation in the device used to measure grip strength on the identification of low muscle strength: Findings from a randomised cross-over study.** Journal of Frailty, Sarcopenia and Falls, 06(04), 225–230. (2021). [\[Link\]](#)
- [10] Du, W., Cornett, K. M. D., Donlevy, G. A., Burns, J., & McKay, M. J. **Variability between Different Hand-Held Dynamometers for Measuring Muscle Strength.** Sensors, 24(6). (2024). [\[Link\]](#)
- [11] King, T. I.. **Inter-Instrument reliability of the Jamar electronic dynamometer and pinch gauge compared with the Jamar hydraulic dynamometer and B&L Engineering mechanical pinch gauge.** The American Journal of Occupational Therapy : Official Publication of the American Occupational Therapy Association, 67(4), 480–483. (2013). [\[Link\]](#)
- [12] Lessof, C., Cooper, R., Wong, A., Bendayan, R., Caleyachetty, R., Cheshire, H., Cosco, T., Elhakeem, A., Hansell, A. L., Kaushal, A., Kuh, D., Martin, D., Minelli, C., Muthuri, S., Popham, M., Shaheen, S. O., Sturgis, P., & Hardy, R. **Comparison of devices used to measure blood pressure, grip strength and lung function: A randomised cross-over study.** PLoS ONE, 18(12 December).(2023). [\[Link\]](#)
- [13] Savas, S., Kilavuz, A., Kayhan Koçak, F. Ö., & Cavdar, S. **Comparison of Grip Strength Measurements by Widely Used Three Dynamometers in Outpatients Aged 60 Years and Over.** Journal of Clinical Medicine, 12(13).(2023). [\[Link\]](#)